

SafeSense

SMART MULTI-HAZARD DETECTION SYSTEM

What is SafeSense?

SafeSense is an Arduino-based safety system that detects earthquakes and fire risks in real time using vibration and temperature sensors. It constantly monitors the environment and immediately shows clear alerts on a display, telling users what danger is happening and what actions to take. If both hazards occur at the same time, it triggers a higher-level emergency warning, helping users respond quickly and stay safe in critical situations.

Problem:

Many people are not alerted early enough during emergencies like earthquakes or fires, and even when they are, they may not know the correct immediate action to take. This delay or confusion can increase risk and harm.

Solution:

A low-cost, real-time detection system that:

- Identifies vibrations (earthquakes)
- Detects high temperature (fire risk)
- Displays clear safety instructions instantly
- Differentiates between single and combined hazards

How It Works:

- A vibration sensor detects movement
- A temperature sensor monitors heat levels
- The Arduino processes the data
- An LCD displays:
- Earthquake → "Drop, Cover, Hold"
- Fire → "Evacuate Calmly"
- Both → "Danger! Exit Immediately"

Target User:

Households in Disaster-Prone Areas:

This system helps families receive early warnings about earthquakes and fire risks in real time. It provides clear safety instructions, helping people react quickly and reduce injury or panic during emergencies.

Schools and Educational Institutions:

In schools, the system acts as a safety demonstration and early warning tool. It helps protect students and staff by quickly detecting hazards and displaying simple instructions like evacuation or “drop and cover” during earthquakes.

Small Businesses and Offices

For workplaces, the system improves safety by alerting employees to potential fire or structural risks. It supports faster evacuation decisions, helping reduce damage, confusion, and risk during emergencies.

Emergency Preparedness & Learning Environments

This system is useful as a teaching tool for STEM and disaster-awareness education. It demonstrates how sensors and smart systems work together to detect hazards and promote safety awareness.

Key Features

1. Real-Time Hazard Detection

The system continuously monitors environmental conditions using sensors. It instantly detects sudden vibrations (earthquake activity) and abnormal temperature increases (possible fire risk), ensuring that hazards are identified as soon as they occur.

2. Dual-Condition Monitoring:

Unlike single-purpose systems, this project monitors both vibration and temperature at the same time. It can detect individual hazards separately or identify when both occur together, which indicates a more serious emergency situation.

3. Clear Safety Messages:

The system does not just detect danger—it tells users exactly what to do. It displays simple, action-based instructions such as “Drop, Cover, Hold” for earthquakes or “Evacuate Immediately” for fire risks, making it easy for anyone to understand quickly.

4. Modular and Expandable Design:

The system is built in a modular way, allowing additional sensors such as smoke, gas, or motion detectors to be easily integrated in the future without redesigning the entire system.

Future Improvements:

**Mobile alerts
(SMS or app
notifications)**



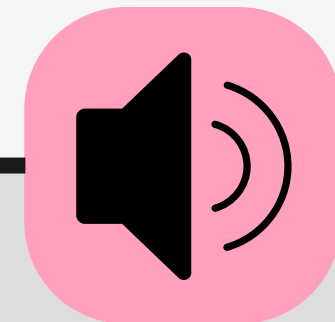
**Louder alarms
and automated
emergency
calls**

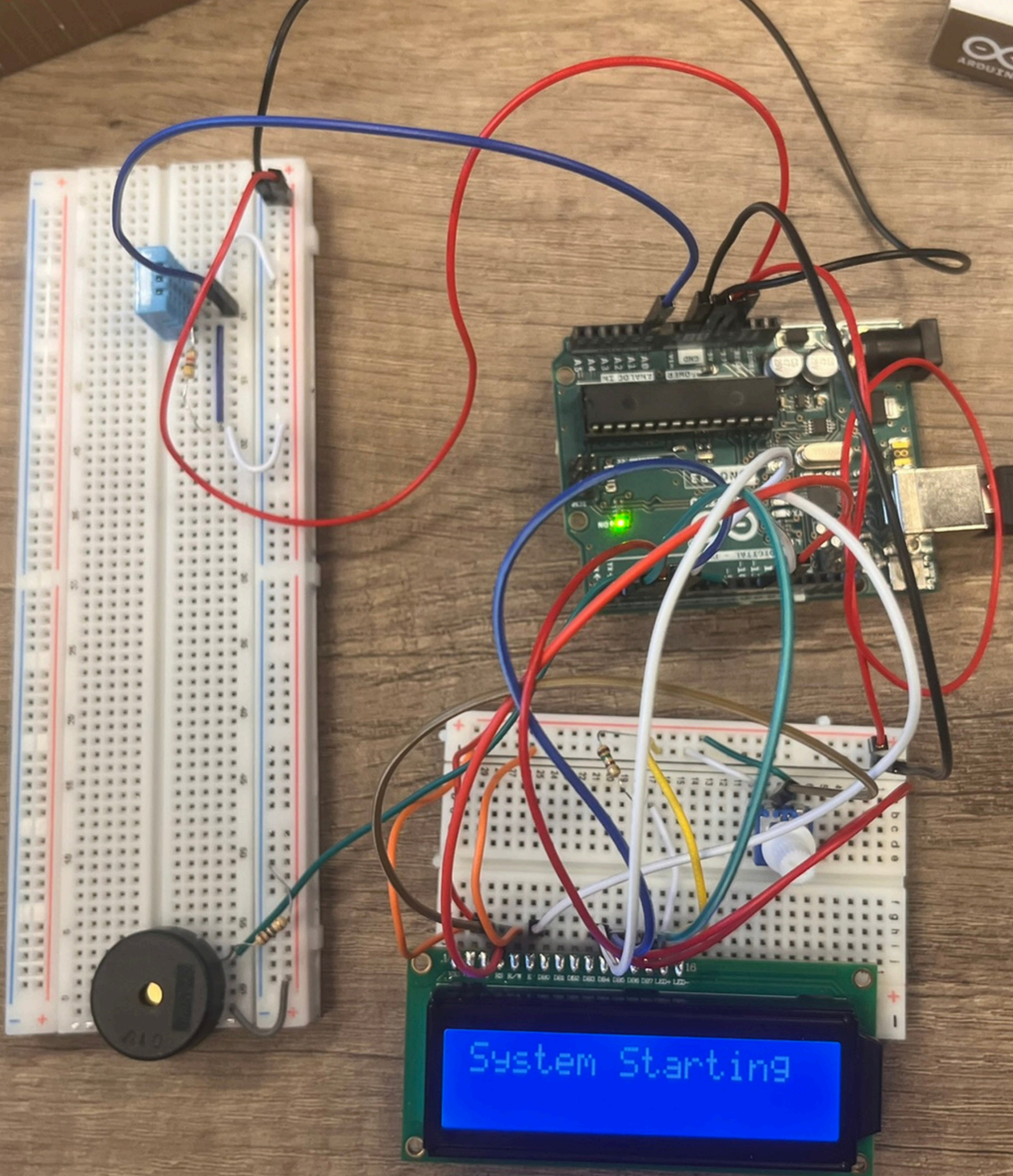


**More sensors
(smoke, gas,
etc.)**



**Voice Alert
System**





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✓ → Arduino Uno

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```
1  #include <LiquidCrystal.h>
2  #include <DHT.h>
3
4  // LCD
5  LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
6
7  // DHT11
8  #define DHTPIN A0
9  #define DHTTYPE DHT11
10 DHT dht(DHTPIN, DHTTYPE);
11
12 // Vibration
13 int VibPin = 8;
14 int VibState = 0;
15
16 // Values
17 float Temp = 0;
18 float Humidity = 0;
19 float TempThreshold = 25.0;
20
21 // Vibration counter (simple filter)
22 int VibCount = 0;
23
24 void setup() {
25     pinMode(VibPin, INPUT_PULLUP);
26     lcd.begin(16, 2);
27     dht.begin();
28
29     lcd.print("System Starting");
```

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```
30     delay(2000);
31     lcd.clear();
32 }
33
34 void loop() {
35
36     // READ DHT11
37     Temp = dht.readTemperature();
38     Humidity = dht.readHumidity();
39
40     if (isnan(Temp)) return; // safety check
41
42     // READ VIBRATION
43     VibState = digitalRead(VibPin);
44
45     // simple noise filter
46     if (VibState == HIGH) {
47         VibCount++;
48     } else {
49         VibCount = 0;
50     }
51
52     if (VibCount > 2) VibCount = 2; // cap value
53
54     // ADAPTIVE TEMP THRESHOLD
55     if (Temp > 25)
56         TempThreshold = 25.0 + (Temp - 35) * 0.2;
57     else
58         TempThreshold = 25.0;
59 }
```

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Arduino Uno



Serial Monitor



safe_sense_main_code.ino



```
59
60 // DISPLAY
61 lcd.clear();
62
63 // CRITICAL ALERT (both)
64 if (VibCount >= 2 && Temp > TempThreshold) {
65     lcd.setCursor(0, 0);
66     lcd.print("CRITICAL ALERT");
67     lcd.setCursor(0, 1);
68     lcd.print("EVACUATE NOW");
69 }
70
71 // EARTHQUAKE ONLY
72 else if (VibCount >= 2) {
73     lcd.setCursor(0, 0);
74     lcd.print("Earthquake");
75     lcd.setCursor(0, 1);
76     lcd.print("Drop & Cover");
77 }
78
79 // FIRE ONLY
80 else if (Temp > TempThreshold) {
81     lcd.setCursor(0, 0);
82     lcd.print("High Temp");
83     lcd.setCursor(0, 1);
84     lcd.print("Fire Risk");
85 }
86
87 // SAFE
88 else {
```

```
// SAFE
else {
    lcd.setCursor(0, 0);
    lcd.print("All Safe");
    lcd.setCursor(0, 1);
    lcd.print("No Danger");
}

delay(1000);
}
```

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Conclusion:

SafeSense demonstrates how simple technology can improve safety and awareness during emergencies. By combining multiple sensors and clear instructions, it helps users respond quickly and effectively, potentially reducing harm in critical situations. This makes the system suitable for real-world early warning applications in homes, schools, and public buildings.

Thank You

FOR YOUR TIME